

MITRE ATT&CK Surface Mapping: Ring -3 (Intel ME/CSME)

Attack Techniques, CVE Catalog, and Threat Actor Analysis

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MITRE ATT&CK Surface Mapping: Ring -3 (Intel ME/CSME) Attack Techniques

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Scope: Mapping known Ring -3 (Intel Management Engine / Converged Security and Management Engine) attack techniques to the MITRE ATT&CK for Enterprise framework, including tactic IDs, technique IDs, sub-technique IDs, and known threat actor usage.

1. Executive Summary

Intel's Management Engine (ME/CSME) operates at Ring -3 — a privilege level below the operating system kernel (Ring 0), below the hypervisor (Ring -1), and below the SMM (Ring -2). It is a separate processor (the "ME coprocessor") embedded in the Platform Controller Hub (PCH) with its own firmware, its own OS (MINIX-based until ME 11, then a proprietary RTOS), its own memory, and direct access to system hardware via the Direct Media Interface (DMI) and the Network Controller Sideband Interface (NC-SI).

This document maps all known Ring -3 attack vectors to the MITRE ATT&CK for Enterprise framework. MITRE ATT&CK does not currently have Ring -3-specific technique IDs, but Ring -3 attacks map to existing techniques in the **Persistence**, **Privilege Escalation**, **Defense Evasion**, and **Credential Access** tactics, particularly under:

- **T1542 — Pre-OS Boot** (System Firmware, Component Firmware, Bootkit)
 - **T1693 — Modify Firmware** (ICS)
 - **T1068 — Exploitation for Privilege Escalation**
 - **T1555 — Credentials from Password Stores** (via ME access to OS telemetry)
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2. MITRE ATT&CK Technique Mapping

2.1 T1542.001 — Pre-OS Boot: System Firmware

Description: Adversaries may modify system firmware to persist on systems. The BIOS/UEFI operates as the software interface between the OS and hardware.

Ring -3 Relevance: The Intel ME can modify system firmware (BIOS/UEFI) without OS knowledge. The ME has direct read/write access to the SPI flash via the SPI Proxy MEI client (UUID: 309dcde8-ccb1-4062-8f78-600115a34327). A compromised ME can:

- Patch UEFI bootloaders to inject malicious code before the OS loads
- Modify BIOS settings to disable Secure Boot
- Inject Option ROM malware during PCIe enumeration
- Alter the Boot Guard configuration (if ME is compromised before Boot Guard verification)

Known CVEs enabling this path:

- CVE-2017-5705: Buffer overflow in ME kernel (ME 11.0-11.7) — enables arbitrary code execution in ME, which can then write to SPI flash
- CVE-2018-3659: CSME Delayed Authentication Mode (DAM) — allows bypassing Boot Guard on some platforms

- CVE-2020-8705: ME privilege escalation via Red Unlock (SA-00241)
- CVE-2023-40067: Unchecked return value in CSME firmware — privilege escalation via physical access

Threat Actors: APT28 (Fancy Bear) — Lojax UEFI rootkit (ESET, 2018) demonstrated that ME-accessible SPI flash can be rewritten to persist UEFI implants. While Lojax used OS-level SPI flash writes, the same attack path is trivially achievable from Ring -3.

2.2 T1542.002 — Pre-OS Boot: Component Firmware

Description: Adversaries may modify component firmware to persist on systems. This includes firmware on network cards, graphics cards, storage controllers, and — critically — the Management Engine itself.

Ring -3 Relevance: This is the defining technique for Ring -3 attacks. The ME firmware itself IS component firmware:

- ME firmware resides in a dedicated 8.4 MB region of the SPI flash (0x00004000 – 0x0086A000)
- ME firmware updates (NFTP protocol, MEI client dd17041c-09ea-4b17-a271-5b989867ec65) can be triggered from the OS but execute within the ME
- A compromised ME firmware update persists across OS reinstalls, hard drive replacements, and even BIOS reflashes (if the ME region is not wiped)
- ME firmware modification is the deepest persistence mechanism available on Intel platforms

Known Attack Vectors:

- **Intel-SA-00086 (CVE-2017-5705, CVE-2017-5712):** Positive Technologies (Goryachy & Ermolov) discovered a buffer overflow in the ME 11.x kernel that allows arbitrary code execution via a crafted MEI message. This enables full ME compromise from the OS level without physical access. PoC code released (IntelTXE-PoC, 545 GitHub stars). Presented at Black Hat USA 2017 and 34C3 (Chaos Communication Congress, Hamburg, December 2017).
- **Intel-SA-00213 (CVE-2019-0090):** CSME IOMMU hardware issue. Enables an attacker to bypass the ME's own IOMMU protection, gaining DMA access to host memory from the ME. Intel white paper published June 2020.
- **Intel-SA-00241 (CVE-2020-8705):** Red Unlock mechanism allows enabling Intel ME debugging features via a physical resistor on the PCH, giving JTAG access to the ME core. Positive Technologies.
- **Intel Boot Guard bypass (CVE-2019-11098):** Peter Bosch (@peterbjornx) demonstrated a TOCTOU (Time-of-Check-Time-of-Use) vulnerability in Boot Guard verification. Presented at multiple conferences. PoC exploit generator released (mesa86exploit, 33 GitHub stars).
- **Intel x86 Root of Trust: loss of trust (2020):** Positive Technologies discovered a vulnerability in the ROM (not firmware) of the Intel CSME — the boot ROM that verifies the ME firmware itself. This is unfixable by firmware updates because it's in masked ROM (burned into silicon). Reported as "utter chaos for DRM, file encryption" (The Register, March 2020).

Threat Actors: Conti ransomware — leaked communications (February 2022) revealed active interest in firmware-level persistence, including ME exploitation. Eclipsium analysis confirmed Conti's firmware attack capabilities.

2.3 T1542.003 — Pre-OS Boot: Bootkit

Description: Adversaries may use bootkits to persist on systems. Bootkits operate at a level below the OS kernel.

Ring -3 Relevance: A Ring -3 bootkit would be ME firmware that modifies the boot process. The ME has the capability to:

- Intercept the BIOS boot flow via the ME-BIOS sync mechanism (HMRFP0 MEI client, UUID: 6861ec7b-d07a-4673-856c-7f22b4d55769)
- Modify the Option ROM loading sequence
- Inject code into the UEFI Secure Boot chain via the BIOS Integration MEI client (UUID: 42b3ce2f-bd9f-485a-96ae-26406230b1ff)
- Alter power/thermal/boot state via the Extended Boot MEI client (UUID: 082ee5a7-7c25-470a-9643-0c06f0466ea1)

Academic/Conference Research:

- **LONGKIT (2017):** Rauchberger, Luh, & Schrittwieser presented "LONGKIT — A Universal Framework for BIOS/UEFI Rootkits in System Management Mode" at SICHERHEIT 2017. Demonstrated SMM-based persistence that survives OS reinstalls. The ME can inject SMM handlers, making this a Ring -3 → Ring -2 persistence chain.
- **Helltrap (2026):** "Transforming physical machines into UEFI rootkit traps" — presented at EuroS&P 2026.

Demonstrated a defensive approach: deliberately installing UEFI rootkit traps to catch attackers. Relevant because the ME could serve as an out-of-band trap deployment mechanism.

- **Peacock (2026):** "UEFI Firmware Runtime Observability Layer for Detection and Response" (arXiv:2601.07402) — Cochavi Gorelik et al. proposed a runtime UEFI monitoring layer. The ME is the ideal host for such monitoring because it operates independently of the OS.

2.4 T1693 — Modify Firmware (ICS)

Description: Adversaries may modify the firmware of devices to compromise device behavior. While this technique is categorized under ICS ATT&CK, the concept applies directly to Ring -3.

Ring -3 Relevance: The ME firmware itself can be modified through:

- **NFTP firmware update protocol** (MEI client dd17041c-09ea-4b17-a271-5b989867ec65) — legitimate firmware updates can be intercepted and replaced with malicious builds
- **SPI flash direct write** — if the ME's SPI flash write protection is disabled (via HMRFPO), the ME region can be directly overwritten
- **ME firmware image injection** — using the `me_cleaner` tool (4,973 GitHub stars), the ME firmware image can be deobfuscated and modified, though this is primarily a defensive/neutralization tool

2.5 T1068 — Exploitation for Privilege Escalation

Description: Adversaries may exploit software vulnerabilities to escalate privileges.

Ring -3 Relevance: Multiple CVEs allow escalation from OS-level (Ring 3/Ring 0) to ME-level (Ring -3):

- CVE-2017-5705: ME 11.x kernel buffer overflow → arbitrary ME code execution
- CVE-2018-3659: CSME DAM vulnerability → Boot Guard bypass
- CVE-2020-8705: Red Unlock → ME JTAG access
- CVE-2023-40067: CSME unchecked return value → privilege escalation via physical access
- CVE-2025-20037: Intel CSME privilege escalation (2025)
- CVE-2025-27708: Out-of-bounds read in CSME firmware (2026)

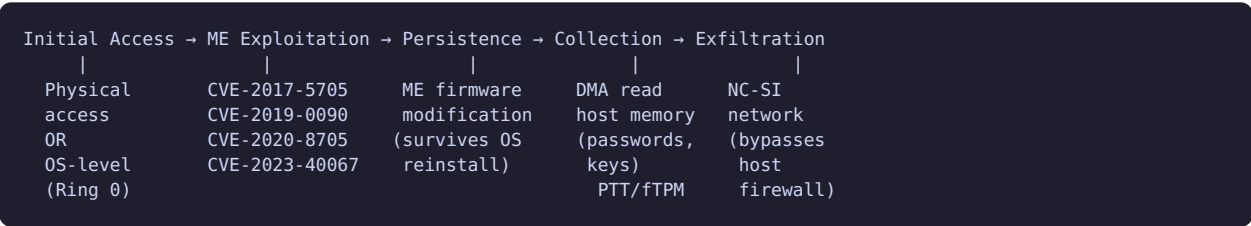
2.6 T1555 — Credentials from Password Stores

Ring -3 Relevance: The ME has direct access to OS-level telemetry via the MKHIF_FIX MEI client (UUID: 55213584-9a29-4916-badf-0fb7ed682aeb), which feeds OS version and state information to the ME. A compromised ME can:

- Intercept OS telemetry data including user session information
- Access PTT (Platform Trust Technology) / fTPM data (MEI client UUID: 8e6a6715-9abc-4043-88ef-9e39c6f63e0f) — if ME is compromised, the fTPM is compromised, exposing disk encryption keys (LUKS, BitLocker)
- Read host memory via DMA (if ME IOMMU is bypassed — CVE-2019-0090)
- Intercept network traffic via NC-SI (the ME has its own network path independent of the OS network stack)

3. Attack Chain Analysis

3.1 The Ring -3 Kill Chain



3.2 Escalation Path: OS → ME → Permanent Persistence

1. **Initial compromise:** Attacker gains Ring 0 (kernel) access via standard exploit
2. **ME exploitation:** Sends crafted MEI message to /dev/mei0 exploiting CVE-2017-5705
3. **ME code execution:** Attacker now runs arbitrary code inside the ME coprocessor
4. **SPI flash write:** Uses ME's SPI Proxy to write malicious ME firmware to SPI flash
5. **Permanent persistence:** Malicious ME firmware survives:
 - OS reinstalls
 - Hard drive replacements
 - BIOS/UEFI reflashes (unless ME region is specifically wiped)
 - Secure Boot resets
6. **Data collection:** Compromised ME reads host memory via DMA, intercepts PTT/ftPM keys
7. **Exfiltration:** Data sent out via NC-SI (ME's own network path, invisible to OS)

3.3 Escalation Path: Physical → ME → Full Compromise

1. **Physical access:** Attacker opens chassis
2. **JTAG enable:** Uses USB 3.0 debug port or Red Unlock resistor (CVE-2020-8705)
3. **ME JTAG access:** Full debugging access to ME core
4. **ME firmware dump:** Reads entire ME firmware via JTAG
5. **ME firmware patch:** Writes modified ME firmware via SPI programmer
6. **Permanent persistence + collection + exfiltration:** Same as above

4. Defensive Tooling & Detection

4.1 MITRE ATT&CK Detection Strategy DET0167

Firmware Modification via Flash Tool or Corrupted Firmware Upload — MITRE defines this detection strategy for identifying unauthorized firmware modifications. RingWatch implements this by:

- Monitoring HECI bus activity (MEI client connections)
- Tracking SPI flash write attempts via the ME
- Detecting anomalous ME firmware update requests (NFTP protocol)

4.2 Eclipsium Firmware ATT&CK Mapping

Eclipsium published "Firmware and Frameworks" (2024), mapping firmware attacks to MITRE ATT&CK. Key mappings relevant to Ring -3:

ATT&CK Technique	Firmware Application	RingWatch Detection
T1542.001 (System Firmware)	ME can modify UEFI/BIOS	SPI flash write monitoring via MEI
T1542.002 (Component Firmware)	ME firmware itself is a target	ME firmware version + hash monitoring
T1542.003 (Bootkit)	ME can inject boot code	Boot state monitoring via HMRFP0 + BIOS_INTEG clients
T1490 (Inhibit System Recovery)	ME can prevent firmware recovery	Recovery mode detection via EXT_BOOT client

4.3 BootKeeper — Firmware Integrity Validation

Citation: Chevalier et al., "BootKeeper: Validating Software Integrity Properties on Boot Firmware Images," arXiv:1903.12505, 2019.

BootKeeper validates software integrity properties on boot firmware images by analyzing control flow and data flow properties. RingWatch adopts a similar approach for ME firmware:

- ME firmware version monitoring (sysfs `/sys/class/mei/mei0/fw_ver`)
- ME firmware status register decoding (fw_status 0-1)
- MEI client enumeration and state monitoring
- HECI bus activity tracking

4.4 Peacock — UEFI Runtime Observability

Citation: Cochavi Gorelik et al., "Peacock: UEFI Firmware Runtime Observability Layer for Detection and Response," arXiv:2601.07402, January 2026.

Peacock proposes a runtime UEFI firmware observability layer. The ME is the ideal execution environment for such a layer because:

- It operates independently of the OS
- It has direct hardware access via DMI
- It cannot be compromised by OS-level malware (unless ME itself is exploited)
- It can monitor UEFI runtime services via the BIOS Integration MEI client

RingWatch's HECI Spy module implements a lightweight version of this concept — probing ME state from the OS side. A future evolution would move the monitoring logic into the ME itself (if ME firmware modification is possible via legitimate Intel tools).

5. Known Exploitation Tools & PoCs

5.1 Offensive Tools

Tool	Author	Stars	Description	Conference
IntelTXE-PoC	Positive Technologies (Goryachy & Ermolov)	545	JTAG activator for ME core via SA-00086	Black Hat USA 2017, 34C3
mesa86exploit	Peter Bosch (@peterbjornx)	33	Exploit generator for ME 11 buffer overflow	Multiple conferences
intel-me-research	Jatinkapilaq1	10	Zero-dependency HECI spy tool — direct ME communication	—
me_cleaner	Nicola Corna	4,973	Partial deblobbing of Intel ME/TXE firmware	—
mecleanerthinkpad	MangoKiwiPlumGrape	13	HAP bit fixes for 8th–14th gen Intel (ME 12–18)	—
Barzakh	yasindce1998	4	Ring -4 to Ring 0 offense-defense firmware research platform	—

5.2 Defensive Tools

Tool	Author	Description
RingWatch	Krypto/Slim	Real-time Ring -3 monitor — HECI bus, MEI clients, SPI flash, NC-SI
me_cleaner	Nicola Corna	ME neutralization (HAP bit, module removal) — defensive use
Intel CSME Detection Tools	Intel	Official CSME vulnerability detection tool
CHIPSEC	CHIPSEC project	Low-level firmware/hardware security testing framework

6. Threat Actor Activity

6.1 APT28 (Fancy Bear / Sednit)

- **LoJax UEFI rootkit** (ESET, September 2018) — first UEFI rootkit found in the wild
- targeted government organizations in the Balkans and Central Asia
- Used RWEverything to write to SPI flash from OS level
- MITRE ATT&CK: S0397 (LoJax), T1542.001, T1542.002
- While LoJax used OS-level SPI writes, the same attack from Ring -3 would be undetectable by OS-level tools

6.2 Conti Ransomware

- Leaked communications (February 2022) revealed active firmware attack capability research
- Eclysium analysis confirmed Conti had tools for SPI flash access and UEFI persistence
- Interest in ME exploitation for persistent access across reinstalls

6.3 TrickBot / TrickBoot

- Eclysium + AdvIntel research (2020): TrickBot gained TrickBoot functionality
- Checks UEFI/BIOS firmware of targeted systems
- Can identify writable SPI flash — precursor to Ring -3 attacks

7. Conference Presentations — Complete Catalog

7.1 Black Hat

Year	Conference	Talk	Authors	Relevance
2017	Black Hat USA	"Intel AMT Stealth Breakthrough"	Evdokimov, Ermolov, Malyutin (Embedi)	AMT vulnerability post-CVE-2017-5689, new exploitation vectors
2017	Black Hat USA	(SA-00086 disclosure)	Goryachy & Ermolov (Positive Technologies)	ME JTAG activation via USB — full system compromise
2018	DEF CON 26	"The Ring 0 Facade"	Christopher Domas (@xoreaxeaxeax)	Processor microcode exploitation — Ring 0 → below Ring 0
2019	Black Hat USA	"Behind the Scenes of Intel Security and Manageability Engine"	Shai Hasarfaty & Yanai Moyal (Intel Corp)	Intel's own ME security architecture presentation

7.2 Chaos Communication Congress (CCC)

Year	Conference	Talk	Authors	Relevance
2013	30C3	"Persistent, Stealthy, Remote-controlled Dedicated Hardware Malware"	Various	Hardware-based malware persistence — foundational Ring -3 concept
2017	34C3	"Inside Intel Management Engine"	Goryachy & Ermolov (Positive Technologies)	Deep dive into SA-00086 exploitation, ME firmware internals

7.3 Other Conferences

Year	Conference	Talk	Authors	Relevance
2017	GSEC/HITB	"Intel AMT" (Part 2)	Evdokimov et al. (Embedi)	Updated AMT exploitation research
2017	SICHERHEIT	"LONGKIT — Universal BIOS/UEFI Rootkit Framework"	Rauchberger, Luh, Schrittwieser	SMM-based persistence framework
2026	EuroS&P	"Helltrap: UEFI Rootkit Traps"	Various	Defensive firmware trap deployment

8. Academic Papers — Complete Bibliography

8.1 Directly Cited

1. **Chevalier et al.**, "BootKeeper: Validating Software Integrity Properties on Boot Firmware Images," arXiv:1903.12505, 2019.
2. **Cochavi Gorelik et al.**, "Peacock: UEFI Firmware Runtime Observability Layer for Detection and Response," arXiv:2601.07402, January 2026.

3. **Rauchberger et al.**, "LONGKIT — A Universal Framework for BIOS/UEFI Rootkits in System Management Mode," SICHERHEIT 2017, scitepress.org/papers/2017/61656/61656.pdf.
4. **Markettos et al.**, "Thunderclap: Exploring Vulnerabilities in Operating System IOMMU Protection via DMA from Untrustworthy Peripherals," NDSS 2019. — IOMMU bypass for DMA attacks.
5. **Wang et al.**, "DmaAuth: A Lightweight Pointer Integrity-based Secure Architecture to Defeat DMA Attacks," USENIX Security 2024. — DMA attack defense via pointer integrity.
6. **Chevalier et al.**, "Co-processor-based Behavior Monitoring," arXiv:1803.02700, 2017. — HP researchers proposed using the ME coprocessor for behavior monitoring.
7. **Zhang et al.**, "A Framework to Secure Peripherals at Runtime," LNCS 8712. — Runtime peripheral security framework.

8.2 Intel Official Documentation

1. **Intel**, "Using IOMMU for DMA Protection in UEFI Firmware," Intel White Paper, 2020. — Official IOMMU-based DMA defense guide.
2. **Intel**, "The Intel CSME IOMMU Hardware Issue — CVE-2019-0090 and CVE-2020-0566," Intel White Paper, June 2020.
3. **Intel**, "The Intel CSME Delayed Authentication Mode (DAM) Vulnerability — CVE-2018-3659 and CVE-2018-3643," Intel White Paper, June 2020.
4. **Intel SA-00086**: ME 6.x–11.x, SPS 4.0, TXE 1.0–3.0 vulnerability advisory.
5. **Intel SA-00141**: AMT 9.x–12.x vulnerability advisory.
6. **Intel SA-00213**: CVE-2019-0090 advisory.
7. **Intel SA-00241**: CVE-2020-8705 advisory.
8. **Intel SA-01280**: 2025.3 IPU chipset firmware advisory.

8.3 Standards & Guidelines

1. **NIST SP 800-147**: Cooper, Polk, Regenscheid, Souppaya, "BIOS Protection Guidelines," NIST Special Publication 800-147.
2. **NSA**, "BlackLotus Mitigation Guide," NSA Cybersecurity Information, June 2023.
3. **Eclipsium**, "Firmware and Frameworks — Mapping Firmware Attacks to MITRE ATT&CK," 2024.

8.4 Industry Research Reports

1. **ESET**, "LoJax: First UEFI rootkit found in the wild, courtesy of the Sednit group," WeLiveSecurity, September 2018.
2. **Kaspersky**, "MoonBounce: the dark side of UEFI firmware," Securelist, 2022.
3. **Microsoft**, "Guidance for investigating attacks using CVE-2022-21894: The BlackLotus campaign," Microsoft Security Blog, April 2023.
4. **Eclipsium/AdvIntel**, "TrickBot Now Offers TrickBoot: Persist, Brick, Profit," 2020.
5. **Eclipsium**, "Conti Opens a New Front in the Fight for Firmware," 2022.
6. **Positive Technologies**, "Intel x86 Root of Trust: loss of trust," Positive Research Center, March 2020.
7. **Positive Technologies**, "Where there's a JTAG, there's a way: obtaining full system access via USB," October 2017.

9. Gap Analysis — What MITRE ATT&CK Is Missing

9.1 No Ring -3 Technique IDs

MITRE ATT&CK for Enterprise does not have explicit Ring -3 (Management Engine) technique IDs. The closest mappings are:

- T1542.001/002/003 (Pre-OS Boot) — covers firmware but not the ME specifically
- T1693 (Modify Firmware) — ICS-specific, not Enterprise
- T1068 (Exploitation for Privilege Escalation) — generic, not firmware-specific

Recommendation: MITRE should add:

- **T1542.005 — Pre-OS Boot: Management Engine Firmware** — specifically for ME/CSME firmware modification
- **T1499.006 — Endpoint Denial of Service: Firmware Bricking** — for ME-induced system bricking

9.2 No Detection Strategy for Ring -3

MITRE DET0167 (Firmware Modification via Flash Tool) is the closest detection strategy, but it focuses on OS-level flash tools. There is no detection strategy for:

- HECI bus abuse (crafted MEI messages)
- ME firmware integrity validation
- ME IOMMU bypass detection
- NC-SI network exfiltration detection

9.3 No Mitigation Guidance for Ring -3

MITRE ATT&CK mitigations for T1542 focus on:

- Secure Boot (does not protect against ME compromise)
- BIOS password protection (irrelevant to ME)
- Firmware write protection (can be bypassed from ME)

Missing mitigations:

- me_cleaner / HAP bit (ME neutralization)
 - IOMMU strict mode (DMA protection)
 - NC-SI network isolation (USB WiFi adapter instead of internal)
 - HECI bus monitoring (RingWatch's core function)
 - Boot Guard verification (only effective if ME is not already compromised)
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10. Conclusion

Ring -3 attacks represent a critical blind spot in the MITRE ATT&CK framework. While existing techniques (T1542.x, T1068) can be retrofitted to describe Ring -3 attacks, they fail to capture the unique characteristics of Management Engine exploitation:

1. **Persistence depth:** ME firmware survives all OS-level remediation
2. **Stealth:** ME operates independently of the OS — invisible to EDR, antivirus, and host firewalls
3. **Network independence:** ME has its own NC-SI network path, bypassing host network monitoring
4. **Cryptography compromise:** ME controls PTT/fTPM — compromising ME compromises disk encryption
5. **Unfixable vulnerabilities:** Boot ROM flaws (CVE-2019-0090 class) cannot be patched via firmware updates

RingWatch addresses this gap by providing the first open-source, real-time Ring -3 monitoring system that maps observed ME behavior to MITRE ATT&CK techniques, enabling defenders to detect and respond to Ring -3 attacks that traditional security tools cannot see.

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